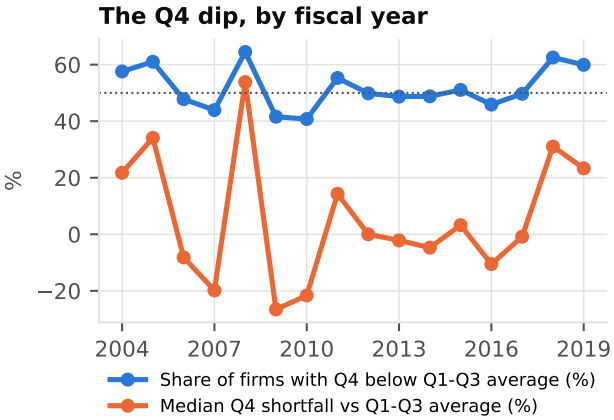
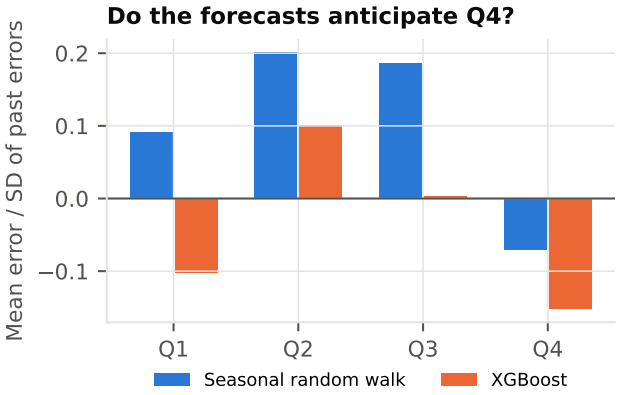
Idea 7 PoC — The fourth-quarter earnings dip

Proof of concept, October 2026. Code: `idea07_q4_dip/run.py`.

Question and design

US firms report fourth-quarter earnings far below the first three quarters (Kapons and Binz), largely because of year-end write-downs and earnings management (Das, Shroff and Zhang, 2009). Do Chinese firms show a similar Q4 dip? Do earnings forecasts and prices anticipate it? For 25,442 firm-years (FY2004–2019) with all four standalone quarters, $DIP = (Q4 - \overline{Q1..Q3})/\sigma$, where σ is the SD of the firm's past SRW errors. The tests are: (a) the size and persistence of the dip; (b) the mean scaled SRW and XGBoost forecast errors by fiscal quarter; (c) the size-adjusted return in the annual-report month, regressed on the XGBoost surprise plus either last year's dip (predictable) or the run-rate surprise that a naive extrapolator would perceive. Regressions have month fixed effects, with SEs clustered by month.





Annual-report month	Announcement month	Months +1 to +3
SUE^{XGB}	0.41 (7.4)	0.36 (3.1)
Last year's dip DIP_{t-1} (known in advance)	0.12 (4.9)	0.00 (0.0)
Run-rate surprise, replacing DIP_{t-1}	0.01 (0.2)	−0.15 (−1.7)

% return per unit (t). The SD of DIP_{t-1} is 4.3, so one SD of past dip is worth about 0.5% at the annual report. Persistence of the dip: slope 0.17 ($t = 4.2$).

Findings

- **No structural Q4 dip in China, but an event-driven one.** In normal years about half of firms report Q4 below their Q1–Q3 average. Large dips cluster in bad years: 2004–05, 2008, and 2018–19, the goodwill-impairment wave, when the median Q4 fell 31% short. They also cluster in loss-making firms (mean DIP −0.71 against −0.25), consistent with year-end big baths rather than a calendar effect.
- **The statistical forecasts are not biased for Q4.** Mean scaled errors in Q4 are small and insignificant for both the SRW ($t = -0.4$) and XGBoost ($t = -0.6$). The time-series models have no systematic Q4 problem to correct.
- **Prices under-anticipate habitual dippers.** Holding the XGBoost surprise fixed, firms with a large dip last year earn lower returns in the annual-report month (about 0.5% per SD), with no reversal afterwards. Investors do not behave like naive run-rate extrapolators: the run-rate surprise adds nothing.

Assessment and next steps

The original hypothesis, a predictable seasonal bias in forecasts, is not supported on these data. The surviving result is narrower: a persistent, firm-specific year-end shortfall that prices do not fully anticipate. Without an analyst FY forecast, the central "do analysts anticipate the dip?" test cannot be run. It needs I/B/E/S FY1 consensus at the Q3 report. A more promising reframing is the 2018 goodwill-impairment wave as a big-bath event study. Overall this idea ranks lowest of Section B.